

3(a)	work (done) / time (taken)	B1
3(b)(i)	zero / 0 J	A1
3(b)(ii)	work done = 440×25 $= 1.1 \times 10^4 \text{ J}$	A1
3(c)(i)	$(\Delta)E_{(P)} = mg(\Delta)h$	C1
	$h = 4.8 \times 10^4 / (1700 \times 9.81)$ $= 2.9 \text{ m}$	A1
3(c)(ii)	$\theta = \sin^{-1} (2.9 / 25)$ $= 6.7^\circ$	A1
3(d)	work done = $4.8 \times 10^4 + 1.1 \times 10^4 (= 5.9 \times 10^4 \text{ J})$	C1
	time = $5.9 \times 10^4 / 1.7 \times 10^4$ $= 3.5 \text{ s}$	A1